Development, simulation & implementation of precoding techniques for satellite links

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***Abstract*—Linear precoding has become a cornerstone of MIMO communications, as it enables spatial multiplexing gains that translate directly into higher spectral efficiency. These benefits are particularly valuable in multi-beam high-throughput satellite systems, where aggressive frequency reuse across the narrow spot beams turns the forward link into an interference-limited MU-MIMO channel. In this environment, precoding is crucial for mitigating inter-beam interference. In mobile satellite scenarios, however, the combination of long propagation delays and time-varying channels induced by user mobility introduce complications. More specific, the CSI available during the precoder computation is systematically outdated with respect to the channel experienced at the time of transmission. This work quantifies the impact of the CSI feedback delay on the system performance, revealing a substantial performance loss. To counter this degradation, a low-complexity prediction strategy is proposed. Numerical results show that this scheme recovers a significant fraction of the performance loss under the considered channel model.**

***Index Terms*—MIMO, Precoding Techniques, Mobile Satellite Systems, Outdated CSI**

I. INTRODUCTION

Satellite communications have recently regained considerable attention as a key enabler of ubiquitous broadband connectivity. They provide cost-efficient coverage of large geographical areas, including regions where deploying terrestrial infrastructure is impractical, and are envisioned as an integral component of future 5G-and-beyond networks [1], [2]. To meet the growing demand for throughput, modern satellite systems have evolved from single-beam architectures towards multi-beam systems, in which the coverage area is partitioned into many narrow spot beams. Combined with aggressive frequency reuse across these beams, this multi-beam architecture turns the forward link into an interference-limited multi-user (MU)-multiple-input multiple-output (MIMO) channel, where linear precoding proves itself a powerful tool for managing the resulting interference [3].

The vast majority of the satellite-precoding literature assumes perfect, instantaneous channel state information (CSI) at the gateway (GW). This assumption becomes unrealistic in mobile satellite scenarios, where user terminal (UT) mobility introduces a Doppler spread that makes the channel time-varying. In addition, propagation delays remain large compared to terrestrial systems: even for a LEO constellation, the round-trip time is on the order of a few milliseconds. As a result, the CSI available at the GW when the precoder is computed is systematically outdated with respect to the channel actually experienced at the time of transmission.

Only a limited number of works have explicitly addressed linear precoding for mobile multi-beam satellite systems under outdated CSI [3]. Vázquez et al. [4] studied delayed CSI in a GEO mobile interactive system. Joroughi et al. [5] proposed robust precoding strategies for a maritime multi-beam scenario. In contrast, the present work starts from a generic multi-user MIMO downlink formulation and studies the effect of outdated CSI on standard linear precoders. Rather than modifying the gateway-side precoder structure, we seek to compensate the mismatch at its source.

The contributions of this paper are twofold. First, the impact of outdated CSI on the achievable sum-rate of three linear precoders, namely zero-forcing (ZF), block diagonalization (BD), and weighted minimum mean square error (WMMSE), is quantified as a function of the feedback delay. Second, a low-complexity compensation strategy based on element-wise autoregressive channel prediction is integrated into the feedback loop and evaluated by means of numerical simulations.

The remainder of the paper is organized as follows. Section II presents the system and channel model. Section III summarizes the considered linear precoders. Section IV introduces the proposed channel-prediction strategy for outdated CSI compensation. Section V reports the simulation results. Section VI outlines directions for future work, and Section VII concludes this work.

II. SYSTEM DESCRIPTION

We consider the forward link of a multi-beam high-throughput satellite (HTS) system. The ground GW centralizes all baseband processing and communicates with K UTs through the satellite, which is equipped with N_T antenna feeds. Each UT has N_R receive antennas, and the system is modeled as a MU-MIMO downlink channel.

System Model. The GW assigns a dedicated precoding matrix $\mathbf{F}_k(t) \in \mathbb{C}^{N_T \times N_S}$ to the N_S-dimensional data symbol vector $\mathbf{a}_k(t)$ intended for UT k. Defining the compound precoder $\mathbf{F}(t) = [\mathbf{F}_1(t) \ \cdots \ \mathbf{F}_K(t)] \in \mathbb{C}^{N_T \times KN_S}$ and the stacked symbol vector $\mathbf{a}(t) = [\mathbf{a}_1^T(t) \ \cdots \ \mathbf{a}_K^T(t)]^T$, the received signal at UT k is

$$\mathbf{y}_k(t) = \mathbf{H}_k(t)\,\mathbf{F}(t)\,\mathbf{a}(t) + \mathbf{n}_k(t), \tag{1}$$

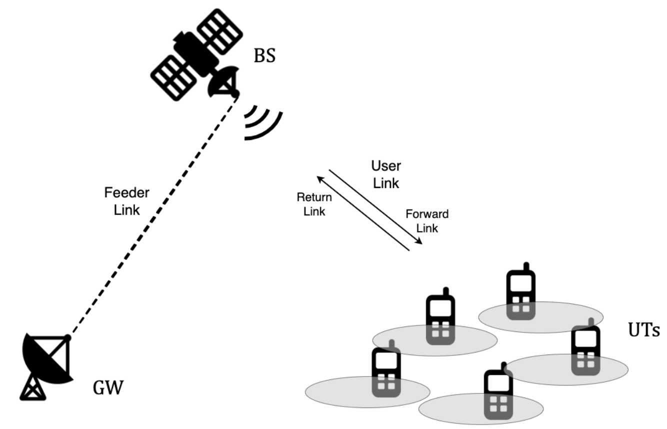


Fig. 1. Illustration of the considered system architecture for mobile satellite communications.

where $\mathbf{H}_k(t) \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix and $\mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, N_0\mathbf{I}_{N_R})$ is additive white Gaussian noise. The data symbols are assumed zero-mean and spatially uncorrelated, and the total transmit power satisfies $\mathrm{tr}(\mathbf{F}(t)\mathbf{F}^H(t)) \leq P_T$ at any moment in time.

Channel Model. The user link of UT k follows a frequency-flat Rician fading model [6], expressed as:

$$\mathbf{H}_k(t) = \sqrt{\frac{\kappa}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{LoS}} + \sqrt{\frac{1}{\kappa+1}}\,\mathbf{H}_k^{\mathrm{NLoS}}(t), \qquad (2)$$

where κ is the Rician factor. The deterministic line-of-sight (LoS) component is treated as time-invariant and modeled as a rank-one matrix, in which all entries share a common random phase offset. The non-line-of-sight (NLoS) component $\mathbf{H}_k^{\mathrm{NLoS}}(t)$ captures the Doppler-induced fading due to UT mobility. Its entries are modeled as i.i.d. circularly symmetric (CS) zero-mean unit-variance complex Gaussian random processes whose temporal correlation follows Jakes' isotropic scattering model. Specifically, the power spectral density of each NLoS channel coefficient is given by

$$S_{h^{\mathrm{NLoS}}}(f) = \frac{1}{\pi f_D \sqrt{1-(f/f_D)^2}}, \quad |f| < f_D, \qquad (3)$$

where $f_D = v f_c / c$ is the maximum Doppler frequency, set by the UT velocity v, the carrier frequency f_c, and the speed of light c. The corresponding autocorrelation function is equal to the zero-order Bessel function of the first kind: $R_{h^{\mathrm{NLoS}}}(\tau) = J_0(2\pi f_D \tau)$.

CSI Feedback Delay. Before the GW can compute the precoder, channel estimates are acquired at the UTs and fed back over the return link. Denoting the total elapsed time as the CSI feedback delay τ_{CSI}, the precoder at time t is designed from the outdated channel state $\mathbf{H}(t-\tau_{\mathrm{CSI}})$. The degree of mismatch is governed by the NLoS coherence time, defined as the lag at which the autocorrelation first falls to one half:

$$T_c^{\mathrm{NLoS}} \triangleq \max\left\{\tau > 0 : R_{h^{\mathrm{NLoS}}}(\tau) \geq \tfrac{1}{2}\right\}. \qquad (4)$$

Throughout this work, results are reported as a function of the normalized delay $\tau_{\mathrm{CSI}}/T_c^{\mathrm{NLoS}}$. This normalization renders the conclusions independent of any specific combination of carrier frequency, UT velocity, or orbital altitude.

III. Precoder Design

Linear precoding designs the transmitted signal in order to limit the interference and efficiently deliver the desired data streams at each UT, by exploiting the available CSI and the spatial degrees of freedom offered by the multi-antenna system. For the system model introduced in the previous section, the design objective is the maximization of the achievable weighted sum-rate (WSR):

$$\max_{\{\mathbf{F}_k\}} \sum_{k=1}^{K} w_k R_k \qquad \text{s.t.} \qquad \sum_{k=1}^{K} \mathrm{tr}\big(\mathbf{F}_k\mathbf{F}_k^H\big) \leq P_T, \qquad (5)$$

where $w_k \geq 0$ is the weight assigned to user k and R_k denotes its achievable rate. In this work, equal weights are used for all users, so that the problem reduces to sum-rate (SR) maximization. Three linear precoders are considered: ZF, BD, and WMMSE.

Zero-Forcing Precoding [7]. ZF precoding cancels all inter-user and inter-stream interference by choosing the compound precoder as the right pseudo-inverse of the compound channel matrix. As a result, each transmitted stream is received over an independent virtual scalar Gaussian subchannel, which allows simple symbol-by-symbol detection. Transmit power is then allocated across these effective subchannels according to the well-known waterfilling solution [8]. The main advantage of ZF is its low complexity and transparent structure. However, when the channel directions are strongly correlated, the required channel inversion becomes inefficient, as lots of power is needed to maintain interference cancellation.

Block Diagonalization Precoding [9]. BD relaxes the ZF constraint by suppressing only the inter-user interference. For each user, the precoder is restricted to the null space of the channel matrices of all other users, such that the multi-user channel is decomposed into parallel K single-user MIMO subchannels. Each effective single-user (SU) channel is then diagonalized by means of its singular value decomposition (SVD), after which waterfilling is applied across the resulting eigenmodes. Compared with ZF, BD exploits the receive-side spatial processing more efficiently and achieves a higher sum rate, especially when each user is equipped with multiple receive antennas.

WMMSE Precoding [10]. Unlike ZF and BD, WMMSE does not enforce hard interference-nulling constraints. Instead, it targets the WSR objective directly through an equivalent weighted minimum mean square error formulation, hence the name. The resulting problem is solved iteratively by alternating between three closed-form updates: the receive combiners, the MSE weights, and the transmit precoder. Among the considered linear precoders, WMMSE provides the best SR performance and serves as the main performance benchmark in the remainder of the paper.

The next sections investigate how these designs degrade when the available CSI is outdated and to what extent this degradation can be mitigated through channel prediction.

IV. Outdated CSI Compensation

The precoders described in the previous section are designed under the implicit assumption that the CSI available at the GW matches the channel experienced at the moment of transmission. In a mobile satellite system, however, this assumption is violated by the combination of a time-varying channel due to UT mobility and the non-negligible feedback delay. As a result, the precoder is computed from an outdated channel realization and is therefore no longer matched to the instantaneous channel. To mitigate this mismatch, a low-complexity channel-prediction strategy is adopted at the UTs.

The key idea is to exploit the temporal correlation of the fading process. Instead of feeding back the most recent channel estimate, each UT predicts the channel that will be experienced after the feedback delay and reports this predicted channel to the GW. The precoder computation itself is left unchanged: only the CSI fed back to the GW is replaced by a prediction. Under the simplified Rician model of Section II, the time variation is entirely contained in the NLoS component, whose entries are spatially uncorrelated and share the same temporal correlation function. For this reason, prediction is performed element-wise using the same scalar predictor for each channel coefficient.

A pth-order autoregressive predictor is used. Let $\hat{h}_{k,(m,n)}[j]$ denote the estimate of the (m,n)-th coefficient of the channel matrix of user k at the j-th pilot instant. The predicted coefficient Δ pilot intervals ahead is then written as

$$\hat{h}_{k,(m,n)}[j+\Delta] = \sum_{\ell=0}^{p-1} \alpha_\ell \, \hat{h}_{k,(m,n)}[j-\ell], \tag{6}$$

where the prediction horizon is chosen such that $\Delta T_p = \tau_{\mathrm{CSI}}$, with T_p the pilot period. The predictor coefficients $\{\alpha_\ell\}$ are obtained from the Yule-Walker equations associated with the known channel autocorrelation function of the entire channel:

$$\mathbf{R}_h \, \boldsymbol{\alpha} = \mathbf{r}_h, \tag{7}$$

where the Toeplitz matrix $\mathbf{R}_h$ and vector $\mathbf{r}_h$ are constructed from $R_h(\tau) = \frac{\kappa}{\kappa+1} + \frac{1}{\kappa+1} J_0(2\pi f_D \tau)$. Since this autocorrelation is fully determined by the maximum Doppler frequency, the predictor coefficients can be computed offline once the model parameters have been selected.

The predictor is governed by two design parameters: the pilot period T_p, which sets the temporal resolution of the channel observations, and the context window $L \triangleq pT_p$, which controls how far back in time the predictor reaches. The model order $p = L/T_p$ follows directly. A tuning study, omitted here for brevity, selected a pilot rate of 6 messages per coherence period and a context window equal to one coherence period. These values are used in all subsequent simulations.

Three idealizations underlie the proposed scheme. First, channel-estimation errors at the UTs are neglected. Second, each UT is assumed to know the second-order statistics of its channel, which here reduces to knowledge of f_D. Third, the pilot period T_p is treated as a free design variable rather than being constrained by a pilot-overhead budget.

V. Simulation Results

This section evaluates the proposed framework through numerical Monte Carlo simulations. We first recall the relative performance of the considered linear precoders under ideal CSI, and then focus on the impact of CSI aging and the gain achieved by the proposed prediction strategy. Unless stated otherwise, the results correspond to an HTS system serving $K = 4$ UTs, with $N_T = 8$ feeds at the satellite and $N_R = 2$ receive antennas per UT. The Rician factor is set to $\kappa = 5$ dB, a common value for urban environments.

As a baseline, Fig. 2 compares the achievable sum rate of ZF, BD, and WMMSE precoding under a purely Rayleigh fading channel and perfect instantaneous CSI. BD consistently outperforms ZF by exploiting the receive-side spatial degrees of freedom of each UT, while WMMSE achieves the highest sum rate across the entire signal-to-noise ratio (SNR) range by trading off interference suppression against desired-signal power. The gap between BD and WMMSE disappears at high SNR, where the regularization in the WMMSE update vanishes and the design effectively reduces to BD. In the remainder of this section, the analysis focuses on WMMSE.

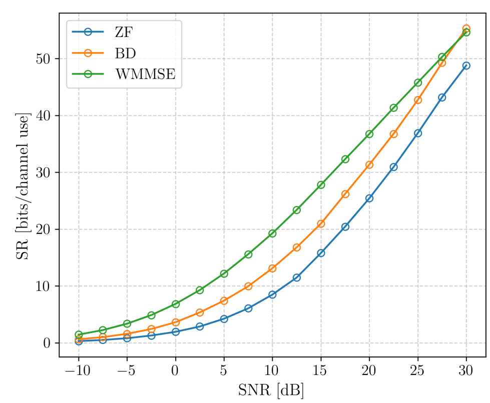


Fig. 2. Comparison of the achievable sum-rate for different precoding schemes under Rayleigh fading and perfect CSI in a $8 \times \{2, 2, 2, 2\}$ system.

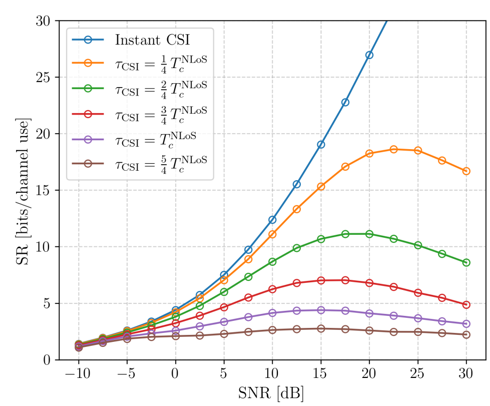
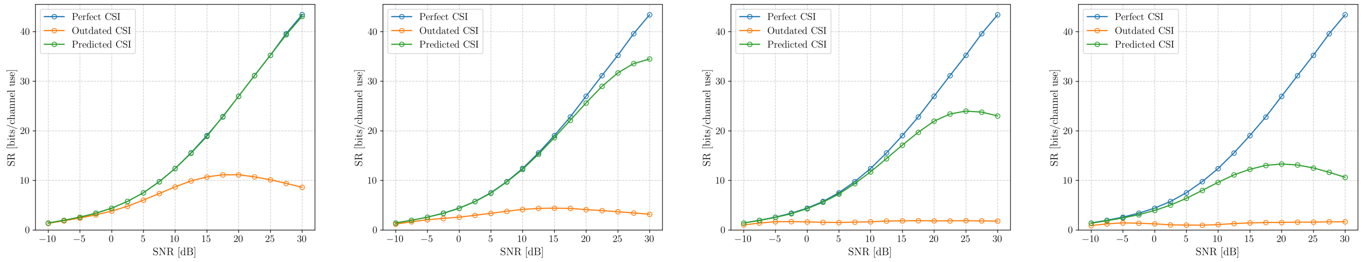


Fig. 3. Comparison of the achievable sum-rate using a WMMSE precoder for different values of the normalized CSI feedback delay.



(a) $\tau_{\text{CSI}} = \frac{1}{2}\,T_c^{\text{NLoS}}$ (b) $\tau_{\text{CSI}} = T_c^{\text{NLoS}}$ (c) $\tau_{\text{CSI}} = \frac{3}{2}\,T_c^{\text{NLoS}}$ (d) $\tau_{\text{CSI}} = 2\,T_c^{\text{NLoS}}$

Fig. 4. Comparison of the achievable sum-rate using a WMMSE precoder with and without the AR-based channel prediction, for different values of the normalized CSI feedback delay.

The conclusions for ZF and BD are in general qualitatively similar but quantitatively more pessimistic.

Fig. 3 reports the sum rate of WMMSE precoding under the Rician channel of Section II for several values of the normalized feedback delay, ranging from instantaneous CSI ($\tau_{\text{CSI}} = 0$) to $\tau_{\text{CSI}} = 5/4\,T_c^{\text{NLoS}}$. Three observations stand out. First, even when the CSI is perfectly instantaneous, the achievable rate lies noticeably below the Rayleigh reference of Fig. 2. This is a direct consequence of the rank-one LoS component of the Rician channel, which concentrates a substantial fraction of the channel energy along a single spatial direction and thereby limits the spatial multiplexing gain. Second, the precoder is highly sensitive to outdated CSI: even a delay of a quarter of the coherence time causes a significant performance loss at moderate-to-high SNR, and the rate degrades monotonically with an increasing feedback delay. Beyond the coherence time, the precoder fails to exploit the spatial dimension of the channel altogether, causing the sum-rate to collapse. The third observation is a characteristic feature of WMMSE precoding with outdated CSI. The sum-rate becomes non-monotonic in function of the SNR, as the precoder increasingly concentrates power along spatial directions that no longer correspond to the desired channel directions and thus actively generates inter-user interference at high SNR.

Fig. 4 evaluates the proposed channel-prediction strategy for four operating points $\tau_{\text{CSI}}/T_c^{\text{NLoS}} \in \{0.5, 1.0, 1.5, 2.0\}$, spanning moderate to severe CSI aging. In each subplot, the sum-rate curve obtained with the autoregressive (AR) predictor at the UTs is compared against the uncompensated case and the perfect-CSI reference. For moderate delays (Figs. 4a and 4b), the predictor recovers most of the performance loss and brings the achievable rate close to the perfect-CSI upper bound across the entire SNR range. The non-monotonic high-SNR behaviour observed without compensation disappears, as the predicted CSI is accurate enough to keep the precoder aligned with the true channel directions. For larger delays (Figs. 4c and 4d), the predictor still provides a substantial improvement, but a residual gap with respect to the perfect-CSI reference becomes visible. Overall, these results show that a low-complexity AR predictor, requiring only knowledge of the channel autocorrelation and no modification of the GW-side precoder, is sufficient to restore a large fraction of the performance lost to outdated CSI in the operating regime of practical interest, under the considered channel model.

VI. Future Work

The simplified Rician model adopted in this paper neglects spatial correlation between channel entries and assumes perfect pre-compensation of the LoS Doppler shift. A natural next step is therefore to extend the analysis to more realistic ray-based satellite channel [11] model that can explicitly capture both effects. First results obtained in the dissertation suggest that the proposed AR-based prediction strategy remains effective in this setting. At the same time, the realistic model reveals an important structural limitation: the channel of each user is effectively rank one, so that each UT can support at most one spatial stream. Future work should investigate the impact of this rank deficiency on different precoding strategies and develop prediction methods that exploit the joint spatial-temporal correlation structure of the channel.

VII. Conclusion

This paper studied the impact of outdated CSI on linear precoding for mobile satellite systems and proposed a low-complexity compensation strategy based on channel prediction at the UTs. Starting from a multi-user MIMO downlink model with Rician fading, three linear precoders were considered: ZF, BD, and WMMSE. The simulations showed that all of them are highly sensitive to CSI aging.

To mitigate this degradation, an element-wise AR predictor was introduced at the UTs. The numerical results showed that this simple strategy restores a large fraction of the performance lost to outdated CSI. For feedback delays up to about one coherence period of the NLoS component, the predicted CSI performance approaches the instantaneous CSI reference very closely. Even for more severe delay values, the prediction remains clearly beneficial and suppresses much of the high-SNR degradation observed without compensation.

A practical example illustrates the relevance of these results. Consider an mobile satellite system (MSS) forward link in the L-band at $f_c = 1.5$ GHz, with a LEO satellite at an altitude of 550 km and a UT velocity of 20 km/h. Since the satellite's trajectory is known with high accuracy, its Doppler shift is deterministic and can be pre-compensated [12]. Under the model adopted in this work, the resulting total CSI feedback delay is approximately $\tau_{\mathrm{CSI}} = 7.34$ ms, while the coherence time of the NLoS component is about $T_c^{\mathrm{NLoS}} = 8.71$ ms. This yields a ratio of $\tau_{\mathrm{CSI}}/T_c^{\mathrm{NLoS}} \approx 0.84$, which lies in the regime where the proposed predictor is most effective.

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